WEEK 5 : CI/CD Pipeline with Azure DevOps

Pre-requisites:

- Ensure **SSH Public Key** or **Personal Access Token (PAT)** for HTTPS is already configured.
- Install **Git**, **Python**, and **VS Code / any IDE**.
- Have your project folder ready on your local system, which includes:
    - Python file for **expense alert**
    - Azure pipeline YAML file

Step 1: Create Python and YAML Files in Local Project

1. Create a Python file (e.g. report_generator.py)
   Example:

```
# report_generator.py

import pandas as pd

# === Load Week-4 cleaned data files ===
students   = pd.read_csv('students.csv')
courses    = pd.read_csv('courses.csv')
enrollments = pd.read_csv('enrollments.csv')
progress   = pd.read_csv('progress.csv')

# === Consolidate the data ===
df = (enrollments
     .merge(students,  on='student_id')
     .merge(courses,   on='course_id')
     .merge(progress,  on=['student_id','course_id'])
     .rename(columns={'progress':'completion_percent'}))

# === Filter for < 50% progress ===
low_progress_df = df[df['completion_percent'] < 50]

# === Save the weekly report ===
low_progress_df.to_csv('weekly_low_progress_report.csv', index=False)
print("Report saved as weekly_low_progress_report.csv")
```

2.Create a YAML file for Azure Pipeline (e.g. azure-pipelines.yml)
Example:

```yaml
trigger:
  branches:
    include:
      - main

schedules:
- cron: "0 9 * * 1"
  displayName: Weekly Report
  branches:
    include:
      - main
  always: true

pool:
  vmImage: 'ubuntu-latest'

steps:
- task: UsePythonVersion@0
  inputs:
    versionSpec: '3.x'

- script: |
    pip install pandas
    python report_generator.py
  displayName: 'Generate Weekly Progress Report'

- task: PublishBuildArtifacts@1
  inputs:
```

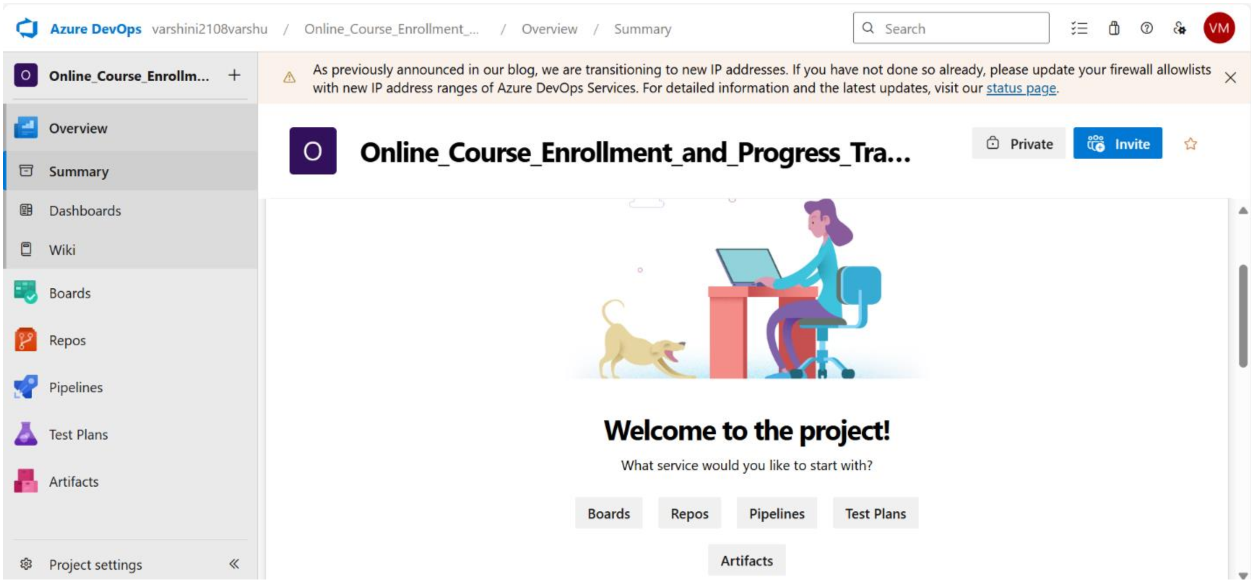
PathtoPublish: 'weekly_low_progress_report.csv'

ArtifactName: 'WeeklyReport'

publishLocation: 'Container'

Step 2: Create a New Azure DevOps Project

1. Go to [Azure DevOps Portal](#).
2. Click **New Project** → Provide name and visibility → Click **Create**.
3. Navigate to **Repos** → Click **Clone** → Copy the **SSH** URL.



Step 3: Push Local Project to Azure Repo via SSH

Open **Command Prompt / Git Bash**, then run the following:

Go to the directory where your local project exists

cd path\to\your\project-folder

Initialize git repository

git init

Add files to git

git add .

Commit the files

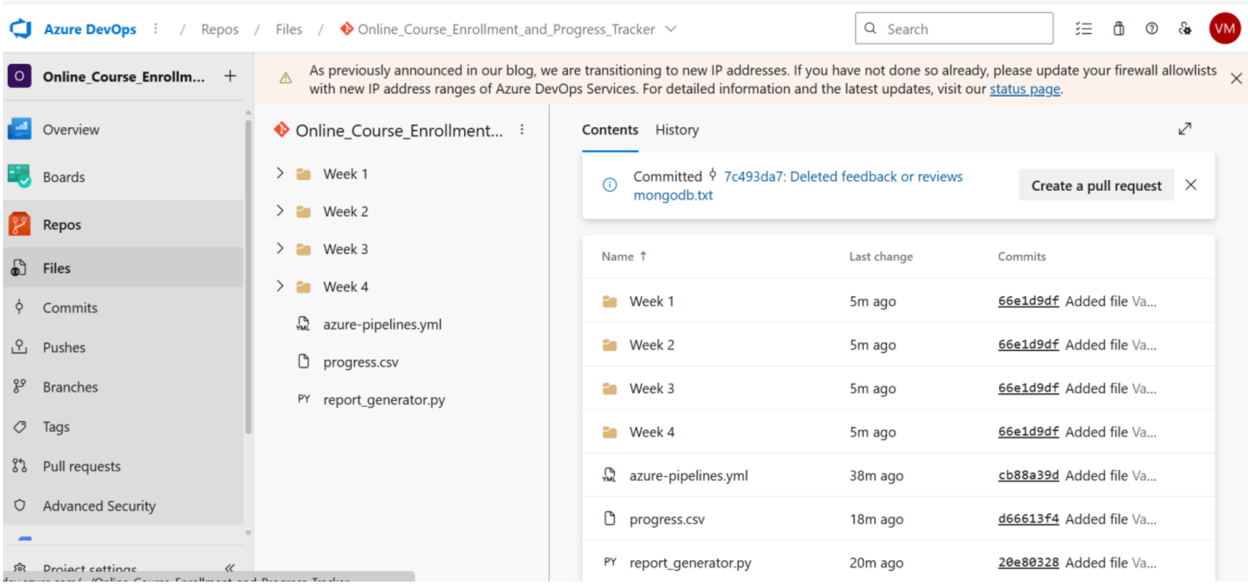
git commit -m "Initial commit"

Add Azure DevOps repo as remote (replace with your SSH link)

git remote add origin git@ssh.dev.azure.com:v3/YourOrg/YourProject/YourRepo

Push the code to Azure repo

git push -u origin main

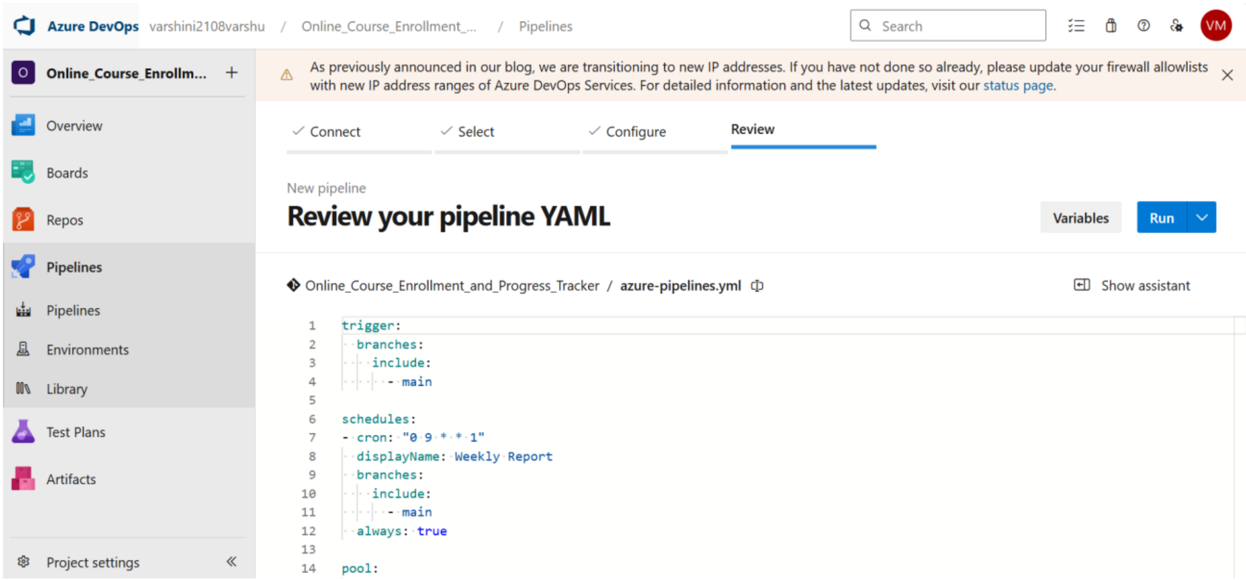


Step 4: Configure and Run the Azure Pipeline

1. Go back to your Azure DevOps project.
2. Navigate to **Pipelines** → Click **Create Pipeline**.
3. Choose:
   - **Code in Azure Repos Git**
   - Select your repository
   - Choose **"Existing Azure Pipelines YAML file"**
4. Select:
   - **Branch:** main

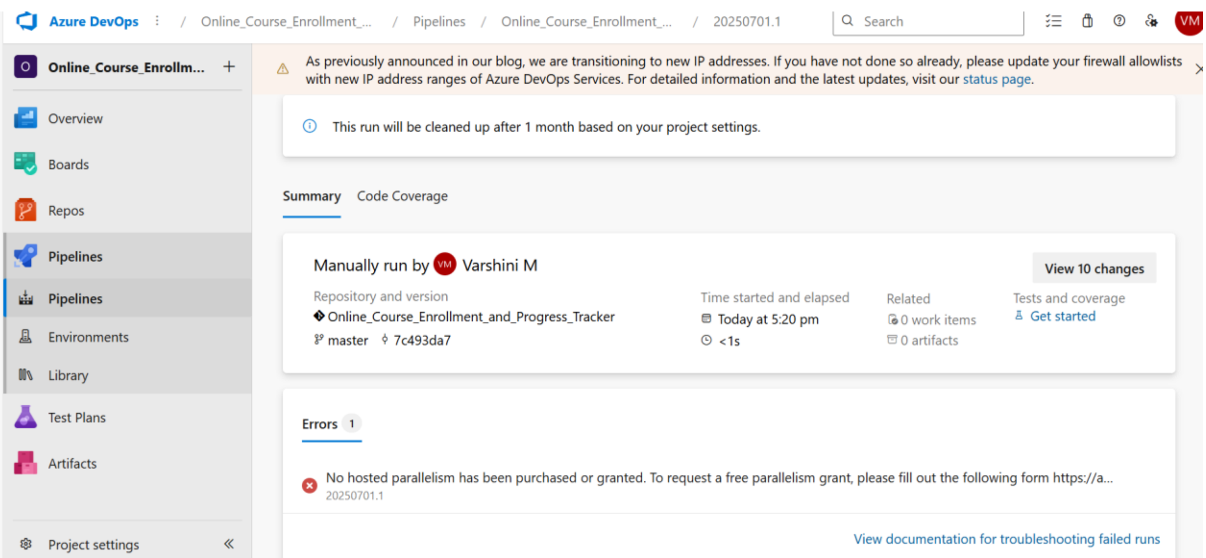
- **Path:** /devops/azure-pipelines.yml

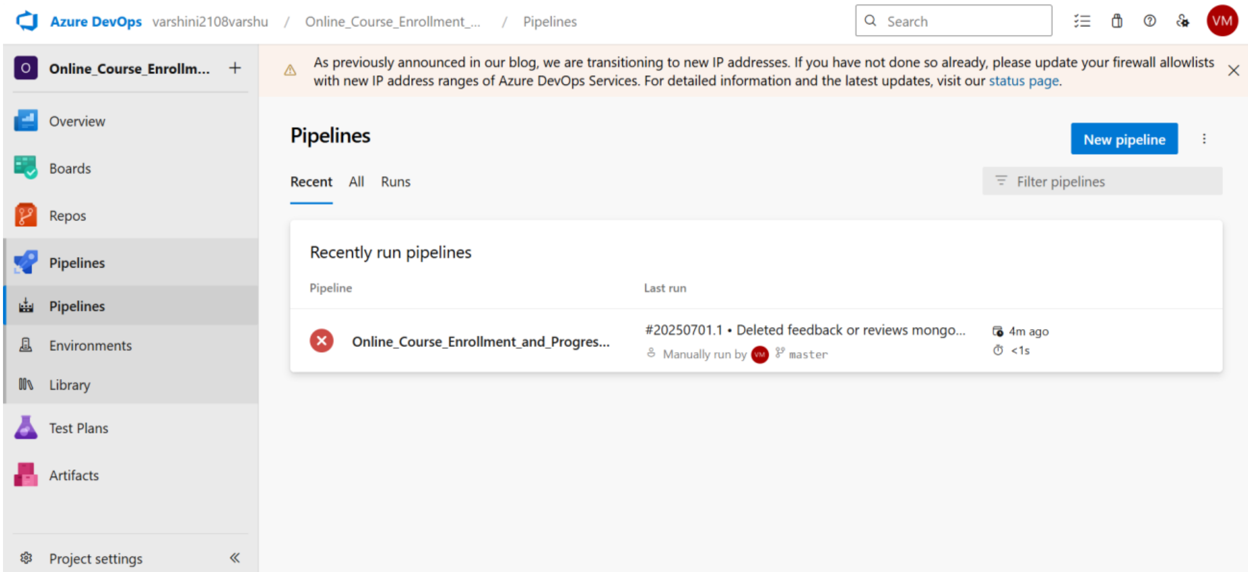
5. Click **Continue**, then **Run** the pipeline.



☑ Final Output

- The pipeline will automatically run every Monday at 9 AM UTC.
- It will generate `weekly_low_progress_report.csv` with students having less than 50% progress.
- The CSV will be available as a build artifact in Azure DevOps.

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As previously announced in our blog, we are transitioning to new IP addresses. If you have not done so already, please update your firewall allowlists with new IP address ranges of Azure DevOps Services. For detailed information and the latest updates, visit our status page.
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